INCH-POUND

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MILITARY SPECIFICATION

ROVING, GLASS, FIBROUS (FOR PREPREG TAPE AN D ROVING, FILAMENT WINDING, AND PULTRUSION APPLICATIONS

This document is inactive for new design after 7 August 2000.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of certain constructions of continuous filament, single or multiple end, fibrous glass rovings. The rovings will be used in prepreg tape and filament winding applications for reinforced plastic structures.

1.2 Classification. The rovings will be of the following types, filament sizes, classes, and yields as specified (see 6.2, 6.3, and 6.8).

Type I – "E" glass

G, H, J, K, L, M, N, P, S, and T filaments

Class 1 – Sizing system compatible with epoxy resin.
Class 2 – Sizing system compatible with polyester resin.
Class 3 – Sizing system compatibility as specified.

Type II – (see 6.8)

Type III – High tensil strength "S" glass

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G filament

Class 1 – Sizing system compatible with epoxy resin.

Type IV – General purpose high tensile strength “S” glass.

G filament

Class 1 – Sizing system compatible with epoxy resin.
Class 2 – Sizing system compatible with polyester.

Roving yield (applicable to all types, filament code designations and classes) as specified (see 3.1.4).

2. APPLICABLE DOCUMENTS

This section not used.

3. REQUIREMENTS

3.1 Roving.

3.1.1 Construction. The roving shall consist of a number of continuous glass filaments coated with suitable sizing system (see 3.1.5) and laid essentially parallel with little or no twist. The filaments may or may not be grouped into strands, intermediate in size, between a single filament and the roving itself (see 6.3). The roving shall be supplied in the form of roving balls (see 3.3). The construction shall be in accordance with 3.3, and the acceptable quality levels (AQL), specified in TABLE I (see 4.2.1.1).

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TABLE I. Examination of roving ball for visual defects.

Examine	AQL Percent Defective		Defect[1/]	Method Inspection
Workmanship:				
Outer layer2/ of roving ball (see 1.2, 3.4,	1.0		Dirt, grease, oil or foreign matter in quantity sufficient to	Visual
3.4.1,	1.0		be detrimental to end	Visual
and 6.2)	1.0		use.	Visual
			Any knot or fuzz accumulation.	
	[3/]		Any splice or group of two or more slices not in accordance with 3.4.1.	Visual
	2.5		Difference in color,	Carpenter's
Package (ball) build:			transparency or other appearance of the	square & scale
Roving ball (see 3.3, 6.2, and 6.4)	2.5		filaments.	Carpenter's square & scale
			Not cylindrical with	
	2.5		5/16 inch	Scale
	2.5			Scale and visual.
	2.5		Not centered on tube, where applicable, within 1/8 inch.	Visual
	2.5			Visual
			Diameter or length	
Tube, (see 3.3 and 6.2)	2.5		improper.	Visual
			Direction and angle of winding improper.	
			Number of turns per transverse improper.	
			Inside diameter or length improper.	
			Not suitable, color code improper.	

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1/ The lot shall be evaluated separately in respect to each category or defect.

/2 Unless the external appearance of a ball or supplier's previous history shows interior of the ball may contain defects, a ball shall not be unwound merely for examination for visual defects.

3/The lot shall be rejected if there is a visual difference in color, transparency, or other appearance of the filaments indicating a possible mixing of different types of glass fibers.

3.1.2 Glass. The chemical composition of "E" glass, as percent by weight, shall be within the limits as follows:

B_2O_3	0 – 10
CaO	16 – 25
$A1_2O_3$	12 – 16
SiO_2	52 – 56
MgO	0 – 5
Na_2O and K_2O	0 – 2
TiO_2	0 – 1.5
Fe_2O_3	0 – 0.8
F_2	0 – 1.0

The chemical composition of high tensile strength "S" glass and general purpose high tensil strength "S" glass shall be certified (see 4.2.1.1) as being within the limits accepted for these two "S" glasses (see 1.2, TABLE II, and 6.3).

3.1.3 Filament diameter and code letter of current commercial filaments. The nominal filament diameter range and code letters shall be as follows (see 6.7 for complete list of code letters).

Filament	Diameter range, inches
G	0.00035 – 0.00040
H	.00040 – .00045
J	.00045 – .00050
K	.00050 – .00055
L	.00055 – .00060
M	.00060 – .00065
N	.00065 – .00070
P	.00070 – .00075
S	.00085 – .00090
T	.00090 – .00095

3.1.4 Roving yield (see 4.3.4, 6.3, and 6.6). Each roving ball fabricated by either conventional or non-conventional roving winding processes shall contain a sufficient number of filaments of the specified diameter code (see 3.1.3) to produce the nominal roving yield specified to within ± 8.5 percent (yield to be specified by procuring activity (see 6.2). Certain type I materials produced by other than conventional winding processes shall have a yield tolerance of ± 13 percent when agreed to by suppliers and the procuring activity.

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TABLE II. Property requirements.

PROPERTY	TYPE I	TYPE III	TYPE IV
[1/] Package (ball density (pounds/cubic inch)			
Minimum	.035	.035	.035
Maximum	.075	.065	.065
[1/] Shore "O" package ball hardness	Classes 1, 2, and 3	Class 1	Class 1, and 2
Minimum	50	50	50
Maximum	90	90	90
Catenary, [2/] inches,/50 feet,			
Maximum	1.0	0.5	1.0
Roving yield	See 3.1.4	See 3.1.4	See 3.1.4
Acetone extractable content, percent of ignition loss			
Minimum	NA[3/]	70	NA[3/]
Resistance to fusion	[4/]	[4/]	[4/]
Tension strength, psi,			
Minimum			
Other than G	200,000	NA	NA
G	280,000	500,000	400,000
Horizontal shear strength (after 6-hour boil) psi,			
Minimum			N/A
Other than G	4,500[5/] or 6,500[5/]		[6/]
G	4,500[5/] or 6,500[5/]		
Specific Gravity range (virgin fiber only)	2.52 – 2.61 (floats in solution Specified in 4.3.8)	NA	N/A

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1/ The supplier has the option to perform either the Package (Ball) Density or the Shore "O" Package Hardness test. Only one of the two tests is required.

2/ Unless otherwise specified, the catenary requirement applies only to rovings having a yield of 700 or more yards per pound.

3/ N/A – Not applicable.

4/ The specimens of 4.3.5 shall not fuse during the specified ignition.

5/These values are applicable only when a specific resin system is not specified by the procuring activity (see 6.2). The 4,500 psi minimum applies to polyester type resin matrix and 6,500 psi minimum applies to epoxy type resin matrix. When deemed necessary, the procuring activity will specify a specific resin system and a corresponding minimum value for horizontal shear strength. For class 3 material, a value agreeable to both supplier and procuring activity shall be agreed upon.

6/Applicable only when specified by the procuring activity (see 6.2). Actual values are dependent upon a given resin. Therefore, values shall be agreed upon between the supplier and the procuring activity for the matrix resin which is to be used. Normal values are 8,500 psi for epoxy and 7,000 psi for polyester resin matrices.

3.1.5 Sizing system (see 6.3). Class 1 roving shall have a sizing system selected and applied so as to provide compatibility between the glass and epoxy resin. Class 2 roving shall have a sizing system which is compatible between the glass and polyester resin, and class 3 shall have a sizing system compatible with the resin specified (see 1.2 and 4.2.1.1).

3.2 Property requirements. The roving shall conform to the property value specified in TABLE II, when tested as specified in the applicable procedure of 4.3 (see 4.2.3).

3.3 Package (ball) build (see table I). The roving shall be would with uniform tension on a tubeless package or a heavy duty winding tube of the dimensions and color code specified (see 6.2) to make essentially a cylindrical "roving ball." The dimensions, direction and angle of winding, the number of turns per transverse, and the spacing of the roving, shall be a specified by the procuring activity (see 6.2). When not specified, the manufacturer's standard dimensions, winding pattern, and color code shall be considered acceptable. The roving shall unwind from the ball on commercial filament winding equipment without excessive process interruptions or excessive tension variation.

3.4 Workmanship. The material shall be free from visible dirt, grease, oil, or foreign matter. Any group of knots, strands or fuzz accumulation occurring at any point in the package which is sufficient to cause an appreciable change in the smoothness of the roving shall be cause for rejection unless the defect can be removed without rendering the remainder of the package rejectable (see table I). There shall be no visible differences in color, transparency, or other appearance of the filaments which may indicate mixing of different types of glass fibers for conventional roving products.

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3.4.1 Splices. When insertion of a broken strand by skip knotting or mechanical insertion method is not acceptable, it shall be spliced with cellulose acetate cement, or any mutually acceptable substitute, with a one inch minimum overlap. When two or more ends are spliced simultaneously, splices shall be spaced so as to prevent any overlapping of the spliced areas, with the exception of gang splices. The maximum number of splices shall not exceed an average of one splice per pound of roving.

3.5 Storage (applicable only to certain type II rovings).When type III rovings, which have sizing systems which will deteriorate at room tempera- tures, are supplied, the supplier shall furnish instructions for storage (see 6.5).

4.VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the con- tractor purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance.Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated.

For purposes of sampling, and in the absence of more formal inspection lot definitions, the inspection lot for examination and tests shall consist of all material of the same type, filament code, and class submitted for delivery and one time.

4.2.1 Inspection of materials and components.In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, examined, and tested in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification will govern.

4.2.1.1 Supplier's certification.The supplier shall certify that (a) all roving balls were produced under the same manufacturing process; (b) the glass is "E", or "S", or general purpose high tensile strength "S" as specified (see 1.2 and 3.1.2); (c) the nominal filament diameter corresponds with code letter designation applicable to the strand specified (see 2.1 and 3.1.3); (d) the sizing on the glass is compatible with the type of resin specified; (e) that all splices if used, were made in accordance with 3.4.1.

4.2.2 Inspection of material.

4.2.2.1 Examination of the roving balls.Examination of the roving balls shall be made in accordance with the classification of defects, inspection levels and AQLs, set forth below. The

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lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of one roving ball for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in package build, appearance, workmanship, splices, and storage instructions.The sample unit for this examination specified in table I shall be 1 roving ball. One set of storage instructions shall be supplied for each container of roving balls.

4.2.2.1.2 Examination of the preparation for delivery.An examination shall be made in accordance with table III to determine that packaging, packing and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected put prior to the closing operations. Shipping containers fully prepared for delivery shall be examined for closure defects.

4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the AQL, expressed as defects per 100 units shall be as follows:

Examination of paragraph	Inspection level	AQL
4.2.2.1.1 4.2.2.1.2	II S-2	See TABLE I 2.5

4.2.2.2 Classification of tests.All tests under this specification shall be classified as lot acceptance tests. Lot acceptance tests shall be made on each lot of material and, in conjunction with the above examination, shall be the basis for acceptance or rejection of the lot.

4.2.3 Testing.The material shall be tested for characteristics listed in table II, in accordance with the test methods specified herein. The lot size for the purpose of determining sample size for testing shall be expressed in units of one roving ball. The sample unit shall consist of sufficient material to prepare all required specimens. Testing for hardness, package (ball) density, catenary and roving yield shall require inspection level II with a 1.0 percent AQL as per MIL-STD-105. All other test shall require inspection level S-2 with a 2.5 percent AQL. One averaged value per ball sampled for each specified property shall determine acceptance as outlined in MIL-STD-105.

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TABLE III. Examination of the preparation for delivery.

Examine	Defect
Ball	Type III roving ball not wrapped in polyethylene film. Types I and IV roving ball not wrapped in polyethylene or other material proving adequate protection. End pad or core plug missing or improper, when applicable.
Packaging	Not level specified; no in accordance with contract requirements. Any nonconforming component, component missing, damaged or otherwise defecting affecting serviceability. Inadequate application of components such as: incomplete closure of polyethylene liners.
Quantity of materials	Less than specified or indicate quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Exterior markings omitted, illegible, incorrect, or improper size, location, sequence or method of application, or no in accordance with contract requirements. When special storage conditions are required, containers not properly marked (see 5.3 and 6.5).

4.3 Testing methods.

4.3.1 Specimen conditioning and testing. Unless otherwise specified, specimens for tensile and shear testing shall be conditioned in accordance with procedure A of ASTM D 618. Specimens for other testing shall be conditioned for a minimum of 2 hours in accordance with ASTM D 618. Testing shall be at 73.4 ± 3.6 F (23 ± 2 C) and 50 ± 5 percent relative humidity.

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4.3.2 Hardness. A shore "O" durometer or equivalent shall be used to test as follows: Set the roving ball on its end on a horizontal flat surface. Support the ball with one hand, hold the durometer in the other, and bring the instrument plunger and foot into contact with outermost layer of roving.
Apply such pressure as is required to bring about a constancy in the reading attained on the dial indicator. Make three readings randomly spaced around the circumference and along the length of the ball.

4.3.3 Catenary (Unless otherwise specified, this test is not applicable to roving having a field less than 700 yards per pound). One specimen per roving ball shall be tested and each obtained by pulling roving from the ball until the slack is taken off, then tying a knot in the roving approximately 12 inches from the package. The roving shall be pulled over a spindle and clamped to a strand which is located approximately 50 feet from the spindle. After returning to the roving ball and holding a 50 foot portion of the strand tight, a hard knot shall be tied. A clamp and sufficient weights to give a total of 2 ± 0.1 pounds shall be placed on the hanging end of the roving. The roving shall be grasped lightly at the center of the span (25 feet) and pulled down 1 foot. The roving shall be released. The pull down and release shall be repeated three additional times. The distance between the furthermost separated strands perpendicular to the lengthwise direction shall be measured.

4.3.4 Roving yield. One specimen shall be tested per roving ball sample. The roving yield shall be computed by dividing the weight into measured length resulting in units such as yards/pound (see 6.3).

4.3.5 Acetone extractable content and resistance to fusion. One specimen per roving ball shall be tested for each requirement in table II as applicable. Testing shall be in accordance with ASTM D 2587 with the following changes:

(a) The specimen shall consist of an accurately measured length of roving weighing approximately 10 grams.

(b) Type I roving shall be ignited at a temperature of 1150 ± 45 F (620 ± 25 C). Types III and IV rovings shall be ignited at a temperature of 1500 ± 45 F (815 ± 25

C). The ignited specimen shall be examined for fusion.

4.3.6 Tensile strength. Five specimens per roving ball shall be tested in accordance with ASTM D 2343. Class 1 of types I, III, and IV roving shall be tested on specimens made with resin conforming to type I, class 1 of MIL-R-9300. When testing class 2 roving, the polyester resin to be used shall be as specified by the procuring activity. When testing class 3 roving, the phenolic, silicone or other resin to be used shall be as specified by the procuring activity. In the event of dispute with either class 1, class 2, or class 3 roving, the resin to be used shall be selected by mutual agreement between the supplier and procuring activity.

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4.3.7 Specific gravity range (applicable to type I only).Six specimens of roving, each approximately 1/3 inch (1 centimeter) long, shall be heat cleaned by exposure to a minimum temperature of 1202 F (650 C) for at least 10 minutes. (Note that the temperature of the heat cleaning process will compact the filaments thereby increasing the actual specific gravity of the virgin fibers. Hence, this method is only for identification of type I roving and not a precise measurement of specific gravity of the virgin fiber.) Three heat cleaned specimens shall be dropped directly into a small container such as a 1 - 1/2 inch diameter test tube with a solution of 2.62 specific gravity which was prepared by mixing 100cc of bromoform (specific gravity 2.84) with 19cc of trichloroethylene (specific gravity 1.46) in a well ventilated hood. Prior to use, the solution should be well sealed to prevent density changes due to evaporation. The roving specimens dropped into the solution shall be allowed to wet out completely. Roving with a specific gravity of 2.61 or less shall be checked for the lower end of the specific gravity range using a mixture of 91cc of bromofoam and 28cc of trichloroethylene yielding a specific gravity of 2.519. The roving specimens shall sink.

4.3.9 Package density. One density determination is made for each ball (package) in the sample from the lot being tested. Density is determined by measuring inside diameter to the nearest 1/16 inch with a steel rule, outside diameter to the nearest 1/16 inch with a circumference tape (or steel rule), length to the nearest 1/16 inch with a steel rule and ball (package) weight to the nearest 0.1 pound and calculating the following:

$$D = \frac{1.27 \times \text{weight}}{\text{length}\ (d_o - d_i)^2}$$

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging authority.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Fibrous glass roving covered by this specification is intended for making glass reinforced structures by use of prepreg tape or roving, filament winding, and pultrusion methods.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number and date of this specification.

b. Type, filament diameter code letter, class and roving yield (yards per pound) required (see 1.2, 3.1.3, 3.1.4, 6.3 and 6.7).

c. Change in yield tolerance to □ □ 13 percent when agreed to by supplier and procuring activity for certain type I materials produced by other than conventional winding processes (see 3.1.4).

d. When deemed necessary for type I roving, specific matrix resin system for horizontal shear strength test and corresponding minimum value (see table II, footnote 4).

e. Resin to be used for tensile strength determination of class 2 and class 3 roving (see 4.3.6).

f. Horizontal shear strength test (see table II, footnote 5 and 4.3.7), if required for types III and IV roving, including specific matrix resin to be used and corresponding minimum value.

g. Outside diameter and length of roving ball (package) (see 3.3 and 6.4).

h. Inside diameter, and length of winding tube (see 3.3).

i. Color coding and dimensions of winding tube, if required (see 3.3).

j. Package (ball) build (winds, direction, turns and spacing), if required (see 3.3).

k. Selection of applicable level of preservation, packaging, and packing required (see 5.1.1.1 and 5.1.1.2).

l. Quantities of roving balls (packages) required in the unit and intermediate packages (see 5.1.1.1 and 5.1.1.2).

m. Special marking, if required (see 3.3).

6.3 Definitions. For the purpose of this specification, the following definitions have been used.

Filament - a single glass fiber, as drawn.

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Strand - a plurality of filaments gathered together with little or no twist and bonded with sizing.

Roving - a plurality of strands of filaments gathered together with little or no twist.

Roving yield - the roving yield in yards per pound is obtained as follows:

$$\text{Roving yield Yards per pound} = \frac{453.6 \times \text{length of specimen (yards)}}{\text{weight of specimen (grams)}}$$

Upon request, the supplier should provide a value for a yield in grams per kilometer (roving weight in TEX).

Roving designation - a roving ordinarily is designated by a roving yield (yards per pound) and a filament diameter code; i.e., "700G" plus the type and class.

Sizing system - a surface treatment or coating applied to the glass during the fiber forming operation to improve its ability to bond to a laminating resin. Different sizing systems may be necessary for use with different resins.

"E" glass - A low alkali, lime-alumina borosilicate glass noted for its good electrical properties (see 3.1.2 for chemical composition).

"S" glass and general purpose high tensile strength "S" glass - A magnesia-alumina-silicate glass of higher tensile strength and melting point than "E".

6.4 Roving ball (Package) dimensions. Ball (package) dimensions should be based on agreement between supplier and purchaser. The following nominal dimensions are being used for the ball (package).

Nominal dimensions inches			Approximate weight, 1/
O.D.	length	I.D. of core	pounds
7	10	3	15
10	10	3	35-40
10	10	6	35
12	10	6	45
14	10	3	77
21	24	6	350-360

1/ The weight will vary with the type of roving.

Commercial packaging and packing of the ball (package) should be based on agreement

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between supplier and purchaser.

6.5 Storage. Certain type III rovings should be stored at subfreezing temperature to prevent deterioration of the sizing. Other type III rovings have stable sizing which do not require refrigeration.6.6 Construction. In the past, it was common practice to attenuate glass through bushings having 204 or 408 holes to produce strands containing those numbers of filaments and then to supply 1, 12, 15, 20, 30 and 60–end rovings made from those strands as described in 6.3. Although many suppliers still are using this practice, others are attenuating fiber glass through bushings containing up to several thousand holes so that they can make heavy roving containing only a single strand. Since the strands are not twisted, rovings made with the larger single strands are equivalent, for most purposes, to those made with the smaller multiple strands. Since existing drawings may specify roving in accordance with the old multiple strand designation, a purchasing activity receiving a requisition for roving having a specific strand yield and end count should check with the requisitioner to determine whether specification, may be considered for substitution.

6.7 Complete list of filament diameters and code letters. The filament diameter range and code letter G through T are as follows:

Filament	Diameter range, inches
G	0.00035 – 0.00040
H	.00040 – .00045
J	.00040 – .00045
K	.00040 – .00045
L	.00040 – .00045
M	.00040 – .00045
N	.00040 – .00045
P	.00040 – .00045
Q	.00040 – .00045
R	.00040 – .00045
S	.00040 – .00045
T	.00040 – .00045

6.8 Type II glass. Type II glass was deleted because it is no longer produced commercially.

6.9 Preservation and packaging.Preservation and packaging will be level A or C, as specified (see 6.2).

6.9.1 Level A.

6.9.1.1 Unit packaging. Unless otherwise specified in the contract or purchase order (see 6.2) roving balls will be packaged in quantities specified by the procuring activity per the method III

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of MIL-P-116. Each ball of type III roving should be wrapped in polyethylene or other material providing adequate protection. Except for type I roving balls, a rigid pad, larger than the roving ball and containing a metal core plug, will be placed against each end of the wrapped roving ball with plugs press fitted tightly into the ball core. The roving ball will be supported by its core in the end pads

6.9.1.2 Intermediate packaging.When required, specified quantities of unit packages will be intermediately packaged as specified in the contract or purchase order (see 6.2).

6.9.2 Level C. Roving balls will be preserved and packaged to afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity. The supplier may use commercially acceptable practice when it meets these requirements.

6.9.3 Packaging. Packing will be level A, B, or C, as specified (see 6.2).

6.9. 3.1 Level A. Roving balls packaged as specified in 5.1.1will be packed in containers conforming to PPP-B-585 (class 3) or PPP-B-601 (overseas type). Unless otherwise specified, containers will be provided with a case liner conforming to MIL-L-10547.Closure and strapping will be per the appendix to the applicable container specification.

6.9.3.2 Level B. Roving balls packaged as specified in 5.1.1 will be packed in containers conforming to PPP-B-585 (class 1), PPP-B-601, or PPP-B-636 (class weather resistant, grade V2s). Closure will be in accordance with the appendix to the applicable container specification.

6.9.3.3 Level C. Roving balls will be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers will be in per the Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

6.9.4 Marking. In addition to any special marking required by the contract or purchase order, containers will be marked in accordance with MIL-STD-129. When special storage conditions are required, containers will be marked to specify completely the conditions for storage (see 6.5) as well as the type of glass.

6.10 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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6.11 Key word.

Continuous filament

Custodians:
Army-MI
Air Force – 20

Preparing Activity
Army – MI
(Project Number 8310-2017-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:30Z
Check the source to verify that this is the current version before use.